

Rishabh Shah

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SUMMARY

Innovative Machine Learning Engineer with a strong focus on neural applications developed at Stanford University. Achieved a 97% cost reduction for motor-impaired communication technology, utilizing advanced algorithms for state-of-the-art speed and accuracy. Seeking to leverage expertise in cutting-edge neural AI to enhance accessibility solutions through technical proficiency and a collaborative mindset.

EXPERIENCE

Clinical Trial Researcher

Stanford University

July 2024 – August 2024, Stanford, CA

- Study in collaboration with Stanford Children's Health and led by Dr. Christopher Lee-Messer.
- Created a clinical trial hosting app for a decentralized study on SYNGAP1-related disorder (<1,500 global cases).
- Led testing of antidepressant Nortriptyline repurposing to improve outcomes for SYNGAP1 patients.
- Built and deployed a global outreach website to recruit and enroll patients worldwide for trial participation.
- Subsidized seizure rates for approx. 20% of patients with disorder.
- Received \$10,000 in funding from the Syngap Research Fund 501(c)(3) nonprofit.

Research Assistant

San Jose State University

July 2024 – August 2024, San Jose, CA

- Developed an inexpensive stress-tracking system using MATLAB and Python for mental health maintenance.
- Prevented panic and anxiety attacks by 17% using k-Nearest-Neighbors ML algorithm and BrainFlow library.
- Produced and directed mental stress time-series neural data collection system for 30+ participants.
- Operated research-grade equipment to produce a live mental stress meter using EEG technology and ML.
- Presented to Alameda County Science Fair, placing 4th in Computational Biology and 150+ Chemistry students.

Machine Learning Engineer

Stanford University, School of Medicine

July 2024 – August 2024, Stanford, CA

- Developed neural EEG-based app for affordable hands-free communication to aid motor-impaired individuals.
 - Decreased prices by 97% (\$20,000 to \$500) for industry-standard motor-impaired communication technology.
 - Mentored by the ex-head of ML at Quizlet and chapter lead of NeuroTechX San Francisco to convert to a startup.
 - Programmed using the P300 speller algorithm with novel word-speller system to reach state-of-the-art speed.
 - Published in IEEE Xplore and invited to present at the International on Artificial Intelligence for Neuroscience.
 - Employed Support Vector Machine, Linear Discriminant Analysis, and Synthetic Minority Over-sampling Technique models to achieve industry-leading 85% accuracy and 23 characters per minute.
 - Used the Muse S EEG Headset (\$300) to achieve affordable design.
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EDUCATION

Associate's Degree

Ohlone University • Fremont, CA • 2023 – 2026 • 4.00

- Relevant coursework: Physics 140: Classical Mechanics, Physics 141: Electricity & Magnetism, Spanish 101B.

High School Diploma

Mission San Jose High School • Fremont, CA • 2022 – 2026 • 4.00

- Valedictorian
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INVOLVEMENT

President & Founder

Fremont, CA • Electrical Engineering Club • May 2024 – Present

- Led an initiative introducing middle schoolers to engineering through hands-on Arduino and circuitry lessons.
- Curated bi-weekly curricula & trained instructors, fundraising \$1,500 to purchase learning kits for school club.
- Expanded club to 40+ members and hosted guest lectures featuring Marvell engineers and UCSC PhD fellows.
- Host weekly Arduino labs at school, increasing access to hardware development to peers.

Founder & CEO

Fremont, CA • Voices of South Asia • April 2024 – present

- Conrad Challenge Innovator Award Cyber-technology and security (top ~3.4%).
- Crowdsourced indigenous natural language data with 500+ volunteers to improve latest chatbot models by 16%.
- Developed website and fine-tuned LLM using collected data. Used data to create robotics courses for 2000+ users.
- Collaborated with local language schools to reach underserved students speaking low-resource languages.
- Reached students across 5 countries with a culturally-aware chatbot version of Gemini and ChatGPT.

President

Fremont, CA • Artificial Intelligence Club • April 2024 – Present

- Led biweekly lectures with slideshows on various AI topics—neural networks and image classification.
- Introduced field-specific applications of AI—ML in agriculture, signal processing & neurotechnology, and more.
- Published handouts giving detailed mathematical explanations of AI concepts to 60+ club members.
- Trained other officers and hosted club events—MAZE day, advertised to over 150 rising freshmen.

Editor and Treasurer

Fremont, CA • Writer's Block Writing Club • January 2024 – Present

- Peer reviewed essays for new writers and authored 20+ articles on web development, technology, books, and AI.
- Organized writing workshops and competitions focusing on MLA, grammar, and creative events and mad libs.
- Contributed to the Medium publication JavaScript in Plain English for a tech community of 169k followers.

Vice President and founder

Fremont, CA • RISE Tennis Organization • February 2023 – present

- Structured affordable local tennis classes with 10+ vetted coaches for elementary and middle schoolers.
- Fundraised \$15,300 to buy sports equipment for underserved kids, led outreach efforts to grow to 30 students.
- Branched to 4 chapters across 3 states, running food drives, donation drives, classes, and other fundraisers.

Team Member

Fremont, CA • School Tennis Team • February 2023 – present

- Regional athletic league 1st place & 2nd place in Bay Area Junior Tennis Tournament.

Volunteer

Fremont, CA • Jain Center of Northern California • October 2022 – present

- Cooked food and organized drives for homeless shelters.
- Led local hackathon to develop eye-tracking communication prototypes for member with motor disability.
- Gifted prototype to family with daughter with CDKL5 disorder, a condition impairing muscle control.
- Introduced Python programming to middle schoolers through weekend lessons.
- Presented and hosted two annual events to crowds of 700+ people.

AWARDS & HONORS

Conrad Challenge Innovator Award

Conrad Challenge • 2025

- Ranked among top 60 in category (top 3.4% overall).

Blue Ocean Entrepreneurship Award

Blue Ocean Entrepreneurship Competition • 2024

- Top 350 pitches of 12,700 participants.

2nd Place, Behavioral Science

Alameda County Science Fair • 2025

- Accepted to publish/present on IEEE Xplore (11% acceptance)
- At the 2nd International Workshop on AI for Neuroscience.

4th Place, Computational Biology

Alameda County Science Fair • 2025

Diamond Challenge Business Competition

Diamond Challenge • 2024

- 3rd regionally, top 200 internationally.

Presidential Volunteer Service Award Gold

Presidential Volunteer Service • 2025

- For 200+ hours volunteered.

Congressional App Challenge State Qualifier

Congressional App Challenge • 2024
• Top 10 in Congressional district CA-17.

American Invitational Math Exam (AIME) Qualification

American Mathematics Competitions • 2026
• Scored 100.5/150 on 2025 AMC 12 B (top 5%).

USA Computing Olympiad, Silver Division

USA Computing Olympiad • 2026

PUBLICATIONS

SpeakEEG: ML Communication app for Motor-impaired Patients

IEEE Xplore • International on Artificial Intelligence for Neuroscience • 2025

SKILLS

Python, Matlab, Arduino, BrainFlow, JavaScript, Tensorflow, Keras, Pandas, MNE-Python, Kivy
